Type MS · Finding unknown constants in a curve from its gradient

Answer 1 of 7 · Question 4 part a · 4 marks

2021 NOV · Paper 02 · extract 1 of 3 · find part a in this mark scheme extract

4 (a)	$f(x)=2(x-3)^2-5$ $\Rightarrow f(x)=2(x^2-6x+9)-5=2x^2-12x+13$ $p=-12,*\ \ q=13$ **ALT** $\frac{dy}{dx}=4x+p=0$ at $(3,-5)$ $\Rightarrow 4\times 3+p=0\Rightarrow p=-12*$ $-5=2\times 3^2-12\times 3+q\Rightarrow q=13$	M1M1 M1A1 cso [4] M1 M1 M1A1 cso [4]
(b)	Minimum since coefficient of x^2 is positive	B1 [1]
(c)	$\frac{dy}{dx}=4x-12\Rightarrow$ when $x=1$ $\frac{dy}{dx}=-8$ So gradient of normal $=\frac{1}{8}$ When $x=1$ $y=3$ $y-'3'=\frac{1}{'8'}(x-1)$ Eg. $y-3=\frac{1}{8}(x-1)$, $x-8y+23=0$ oe	B1 B1ft M1 M1 A1, A1 [6]
		Total 11 marks

Type MS · Finding unknown constants in a curve from its gradient

Answer 1 of 7 · Question 4 part a · 4 marks

2021 NOV · Paper 02 · extract 2 of 3 · find part a in this mark scheme extract

Part	Mark	Notes
(a)	M1	For using the coordinate of the stationary point and forming the expression: $f(x)=2(x-3)^2-5$ This must be correct for this mark
	M1	Expands their $2(x-3)^2-5$ to form a 3TQ
	M1	For equating coefficients with the given equation to find the value of p and q
	A1	For $p=-12,*\ q=13$ **NB:** The value of p is given in the question.
	ALT – Uses differentiation	
	M1	For a correct differentiation of the expression for y This must be correct for this mark
	M1	For equating their $\frac{dy}{dx}=0$, substituting in $x = 3$ and attempting to find a value for p
	M1	For substituting the given value of p into $y=2x^2+px+q$ using the coordinates $(3,-5)$ to find the value of q. Explicit substitution of the value of p into the equation must be seen if p has not been found using differentiation. [M0M0M1A0 can be scored]
	A1	For $p=-12,*\ q=13$ **NB:** The value of p is given in the question.
(b)	B1	For the correct statement: Minimum since coefficient of x^2 is positive
	ALT	
	B1	For $\frac{d^2y}{dx^2}=4\Rightarrow$ positive, hence minimum. A correct conclusion is required.
(c)	B1	For the value of the gradient when $x = 1$ $[m=-8]$
	B1ft	For the gradient of the normal which must have come from differentiation. Ft their -8
	M1	For the value of y when $x = 1$ using their value of q
	M1	For a complete equation of the line in any form using their normal and their value of y Allow this mark for an equation of the line with $x = 1$ and $y =$ '3' with their normal, provided there is evidence that the normal comes from the negative reciprocal of their value of the tangent. For this mark the gradient of the tangent need not have come from differentiation.
	A1	For the correct equation in any form.
	A1	For the correct equation in the required form or any multiples of the coefficients.

Type MS · Finding unknown constants in a curve from its gradient

Answer 1 of 7 · Question 4 part a · 4 marks

2021 NOV · Paper 02 · extract 3 of 3 · find part a in this mark scheme extract

Question number	Scheme	Marks

Type MS · Finding unknown constants in a curve from its gradient

Answer 2 of 7 · Question 9 part a · 5 marks

2025 JUN · Paper 01 · extract 1 of 4 · find part a in this mark scheme extract

Number		
9(a)	$f'(x)=\lambda(x-2)(3x+1)$	M1
	$f(x)=\int\lambda(x-2)(3x+1)\,dx=\left[\int\lambda\left(3x^2-5x-2\right)\right]dx=\lambda\left(\frac{3x^3}{3}-\frac{5x^2}{2}-2x+k\right)$	M1A1
	At $(2,-9)$ $-9=2(2^3)-5(2^2)-4(2)+c\Rightarrow c=3$	M1
	$a=-5,b=-4,c=3\Rightarrow\left\{f(x)=2x^3-5x^2-4x+3\right\}$ *	A1*cso
		[5]
	ALT $f(x)=2x^3+ax^2+bx+c$ $f'(x)=6x^2+2ax+b$	[M1
	Both $\begin{cases}0=6(2^2)+2(2)a+b\\0=6\times\left(-\frac{1}{3}\right)^2+2\times\left(-\frac{1}{3}\right)\times a+b\end{cases}$	M1A1
	$\begin{cases}72+12a+3b=0 \text{ and } 12-12a+18b=0\Rightarrow a=...,b=...\\\text{At } (2,9)\quad -9=2(2^3)-5(2^2)-4(2)+c\Rightarrow c=...\end{cases}$	M1
	$a=-5,b=-4,c=3\Rightarrow\left\{f(x)=2x^3-5x^2-4x+3\right\}$ *	A1*cso]
(b)(i)	$f(-1)=2(-1)^3-5(-1)^2-4(-1)+3=0$ oe	B1 [1]
(b)(ii)	$x+1\overline{)2x^3-5x^2-4x+3}$ $\quad 2x^2-7x+3$	M1
	$2x^2-7x+3=(2x-1)(x-3)\Rightarrow(x+1)(x-3)(2x-1)$	M1A1 [3]
(c)	$A=\int_{-1}^{\frac{1}{2}}(2x^3-5x^2-4x+3)\,dx+\left\lvert\int_{\frac{1}{2}}^{3}(2x^3-5x^2-4x+3)\,dx\right\rvert$	M1
	$A=\left[\frac{1}{2}x^4-\frac{5}{3}x^3-2x^2+3x\right]_{-1}^{\frac{1}{2}}+\left\lvert\left[\frac{1}{2}x^4-\frac{5}{3}x^3-2x^2+3x\right]_{\frac{1}{2}}^{3}\right\rvert$	M1
	$A=\left\{\left[\frac{1}{2}.\left(\frac{1}{2}\right)^4-\frac{5}{3}.\left(\frac{1}{2}\right)^3-2.\left(\frac{1}{2}\right)^2+3\left(\frac{1}{2}\right)\right]-\left[\frac{1}{2}.(-1)^4-\frac{5}{3}.(-1)^3-2.(-1)^2+3.(-1)\right]\right\}$ $+\left\lvert\left\{\left[\frac{1}{2}.(3)^4-\frac{5}{3}.(3)^3-2.(3)^2+3(3)\right]-\left[\frac{1}{2}.\left(\frac{1}{2}\right)^4-\frac{5}{3}.\left(\frac{1}{2}\right)^3-2.\left(\frac{1}{2}\right)^2+3\left(\frac{1}{2}\right)\right]\right\}\right\rvert$	M1
	$A=\left\{\frac{117}{32}+\left\lvert-\frac{1375}{96}\right\rvert\right\}=\frac{863}{48}$	A1
		[4]
		Total 13 marks

Type MS · Finding unknown constants in a curve from its gradient

Answer 2 of 7 · Question 9 part a · 5 marks

2025 JUN · Paper 01 · extract 2 of 4 · find part a in this mark scheme extract

Part	Mark	Additional Guidance
(a)	M1	For forming an expression for $f'(x)$ using the two turning points given. Students may be working with $\lambda = 1$ at this stage and recognise the need to multiply through later.
	M1	Attempts to integrate, see General Guidance for the definition of an attempt Students may be working with $\lambda = 1$ at this stage and recognise the need to multiply through later.
	A1	Correct integration including +c, Students may be working with $\lambda = 1$ at this stage and recognise the need to multiply through later.
	M1	$\lambda = 2$ is stated or implied, $\lambda k = c$ Substitution of $(2,-9)$ to their $f(x)$ (Does not have to be simplified) or substitution of $\left(-\frac{1}{3}, \frac{100}{27}\right)$ (Does not have to be simplified)
	A1*cso	Obtains the **given** answer with no errors in the working
	ALT - starts with $f(x) = 2x^3 + ax^2 + bx + c$	
	M1	Attempts to differentiate, see General Guidance for the definition of an attempt
	M1	Equates differentiated expression to 0 and substitutes $x = 2$ **and** equates differentiated expression to 0 and substitutes $x = -\frac{1}{3}$
	A1	Two correct equations, simplified or unsimplified
	M1	Attempts to solve two **correct** equations simultaneously (including using calculators) to find a, b **and** substitution of their $a, b, (2,-9)$ to $f(x)$ to find c or substitution of their $a, b, \left(-\frac{1}{3}, \frac{100}{27}\right)$ to $f(x)$ to find c
	A1*cso	Obtains the **given** answer with no errors in the working
(b)(i)	B1	Demonstrating that $x+1$ is a factor $f(-1) = 2(-1)^3 - 5(-1)^2 - 4(-1) + 3 = 0$ (factor theorem) If shown via polynomial division, the division must be fully correct and obtains 0 as the remainder.
(b)(ii)	M1	Dividing by $(x+1)$ to obtain a three-term quadratic quotient: $2x^2 - 7x + ...$
	M1	Attempt to factorise their 3TQ, see General Guidance. Or attempt to solve their 3TQ using the quadratic formula
	A1	Correct three linear factors written in one line : e.g. $(x+1)(x-3)(2x-1)$
	ALT I – equates coefficients	
	M1	For stating $2x^3 - 5x^2 - 4x + 3 = (x+1)(Ax^2 + Bx + C) \Rightarrow$ $2x^3 - 5x^2 - 4x + 3 = Ax^3 + (A+B)x^2 + (B+C)x + C \Rightarrow$ $A = 2, B = -7, C = ...$
	M1	Attempt to factorise their 3TQ, see General Guidance. Or attempt to solve their 3TQ using the quadratic formula
	A1	Correct three linear factors written in one line: e.g. $(x+1)(x-3)(2x-1)$
	ALT II - inspection	
	M1	For finding the quadratic factor. Minimum required is $(x+1)(2x^2 - 7x + C)$
	M1	Attempt to factorise their 3TQ, see General Guidance. Or attempt to solve their 3TQ using the quadratic formula
	A1	Correct three linear factors written in one line: e.g. $(x+1)(x-3)(2x-1)$

Type MS · Finding unknown constants in a curve from its gradient

Answer 2 of 7 · Question 9 part a · 5 marks

2025 JUN · Paper 01 · extract 3 of 4 · find part a in this mark scheme extract

(c)	M1	For identifying the correct limits, uses a correct strategy to find the area above the x-axis and below the x-axis: $\int_{-1}^{\frac{1}{2}}(2x^3-5x^2-4x+3)\,dx+\left\lvert\int_{\frac{1}{2}}^{3}(2x^3-5x^2-4x+3)\,dx\right\rvert$ or $\int_{-1}^{\frac{1}{2}}(2x^3-5x^2-4x+3)\,dx-\int_{\frac{1}{2}}^{3}(2x^3-5x^2-4x+3)\,dx$ or implied by their working
	M1	For an attempt to integrate $2x^3-5x^2-4x+3$ See General Guidance for the definition of an attempt.
	M1	For clear substitution of **correct** limits into any **one** of their integrated expressions (must be a changed expression). Can be implied by **one** of the **correct exact** areas found from a **correctly** integrated expression
	A1	For correctly obtaining $\frac{863}{48}$ or $17\frac{47}{48}$

Type MS · Finding unknown constants in a curve from its gradient

Answer 2 of 7 · Question 9 part a · 5 marks

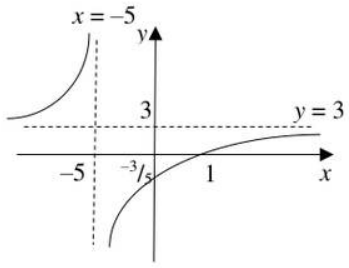
2025 JUN · Paper 01 · extract 4 of 4 · find part a in this mark scheme extract

Question number	Scheme	Marks

Type MS · Finding unknown constants in a curve from its gradient

Answer 3 of 7 · Question 3 part a · 3 marks

2021 JUN · Paper 01 · extract 1 of 3 · find part a in this mark scheme extract

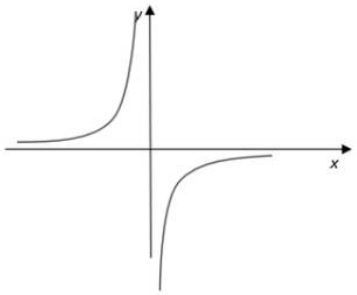
3 (a)	$\frac{dy}{dx} = \frac{a(x+5)-(ax-3)}{(x+5)^2}$	M1
	When $x = 2$ $\frac{dy}{dx} = \frac{7a-2a+3}{49} = \frac{18}{49} \Rightarrow a = ...$	M1
	$a = 3$ *	A1cso (3)
3 (b)(i)	$y = 3$	B1
(b)(ii)	$x = -5$	B1 (2)
(c)(i)	$(1, 0)$	B1
(c)(ii)	$\left(0, \frac{-3}{5}\right)$	B1 (2)
(d)	 Curve drawn Asymptotes drawn and labelled $^{-3}/_{5}$, 1 labelled on axes	B1 B1ft B1ft (3)
		[10]
		Total 10 marks

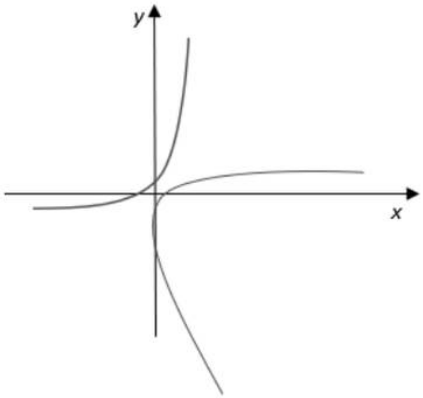
Type MS · Finding unknown constants in a curve from its gradient

Answer 3 of 7 · Question 3 part a · 3 marks

2021 JUN · Paper 01 · extract 2 of 3 · find part a in this mark scheme extract

<table>
<tr><td></td><td></td><td></td></tr>
<tr><td>(b)(i)</td><td>B1</td><td>For $y = 3$ This must be an equation of a line. Do not award for just 3.</td></tr>
<tr><td>(ii)</td><td>B1</td><td>For $x = -5$ This must be the equation of a line. Do not award for just –5</td></tr>
<tr><td>(c)(i)</td><td>B1</td><td>For (1, 0) or clearly listing $x = 1$, $y = 0$ as a pair.</td></tr>
<tr><td>(ii)</td><td>B1</td><td>For $\left(0, -\frac{3}{5}\right)$ oe. or clearly listing $x = 0$, $y = -\frac{3}{5}$ as a pair.</td></tr>
<tr><td rowspan="4">(d)</td><td rowspan="2">B1</td><td>For the curve drawn with two branches anywhere on the grid provided it is a negative reciprocal curve. The ends of the curves must be asymptotic and must not turn back on themselves. Do not allow any obvious overlap across the ends of the curve with evidence of the presence of asymptotes.

For example, accept:
Negative reciprocal curve in incorrect position
 

Do not accept:
• Overlap of asymptotes
• Ends turning back on themselves
 
</td></tr>
<tr><td>B1ft</td><td>For their asymptotes correctly drawn and clearly labelled with their equation.
At least one branch of the curve is required. It must be a negative reciprocal and it must be in the correct position for their asymptotes. The follow through is available for their answers in part b.
If correct asymptotes appear on the sketch, do not award marks retrospective marks in part b.</td></tr>
<tr><td>B1ft</td><td>The curve must be drawn going through their two points of intersection. It must be a negative reciprocal, in the correct position for their intersections and clearly marked on the axes. The follow through is available for their answers in part c. Allow the other branch to even be missing.
If correct coordinates appear on the sketch, do not award marks retrospective marks in part c.</td></tr>
</table>

Type MS · Finding unknown constants in a curve from its gradient

Answer 3 of 7 · Question 3 part a · 3 marks

2021 JUN · Paper 01 · extract 3 of 3 · find part a in this mark scheme extract

Question number	Scheme	Marks

Type MS · Finding unknown constants in a curve from its gradient

Answer 4 of 7 · Question 7 part a · 6 marks

2023 JUN · Paper 01 (regional R) · extract 1 of 2 · find part a in this mark scheme extract

7(a)	$\frac{dy}{dx} = 2mx + \frac{1}{2} \times 64x^{-\frac{1}{2}}$	M1A1
	$32x^{-\frac{1}{2}} + 2mx = 0 \Rightarrow 32 \times 4^{-\frac{1}{2}} + 2m \times 4 = 0 \Rightarrow 16 + 8m = 0 \Rightarrow m = -2$	dM1A1
	$n = 39 + 64\sqrt{4} - 2 \times 4^2 = 135$	dM1A1 [6]
(b)	$\frac{d^2y}{dx^2} = -\frac{1}{2} \times 32x^{-\frac{3}{2}} - 4 = -16x^{-\frac{3}{2}} - 4$	M1
	$\frac{d^2y}{dx^2} = -6 < 0 \Rightarrow$ maximum	A1FT [2]
		Total 8 marks

Type MS · Finding unknown constants in a curve from its gradient

Answer 4 of 7 · Question 7 part a · 6 marks

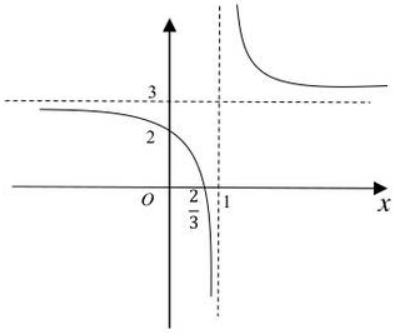
2023 JUN · Paper 01 (regional R) · extract 2 of 2 · find part a in this mark scheme extract

Question	Scheme	Marks

Type MS · Finding unknown constants in a curve from its gradient

Answer 5 of 7 · Question 9 part a.i · 0 marks

2020 JAN · Paper 02 · extract 1 of 3 · find part a.i in this mark scheme extract

9(a)(i)	$\frac{2}{p} = 2$ so $p = 1$ *	M1A1cso
(ii)	$\frac{dy}{dx} = \frac{q(x-1)-(qx-2)}{(x-1)^2}$	M1A1
	When $x = 0$ $\frac{dy}{dx} = \frac{-q+2}{1} = -1, \Rightarrow q = 3$	M1A1,A1 (7)
(b)		B1ft B1ft B1 B1ft B1ft (5)
(c)	$x + 2 = \frac{3x-2}{x-1}$	M1
	$x^2 - 2x = 0$	M1
	$x(x-2) = 0$ $x = 2$	dM1A1cao(4) **[16]**

(a)(i)M1	Set $x = 0$ in the curve equation and equate result to 2. Obtain a value for p.
A1cso	Correct value of p obtained from a correct equation.
(ii)M1	Attempt the quotient rule. (formula is given on formula page). Denominator must be $(x-1)^2$. Numerator to be $q(x-1)-(qx-2)$ or $(qx-2)-q(x-1)$ **Must use $p = 1$ now or later.**
A1	Fully correct derivative
ALT	Use product rule: $\frac{dy}{dx} = q(x-1)^{-1} - (qx-2)(x-1)^{-2}$ M1 for attempt with 2 terms similar to above, either term to be correct A1 Both terms correct
M1	Set $x = 0$ in their derivative and equate to -1
A1	Correct equation
A1	$q = 3$

Type MS · Finding unknown constants in a curve from its gradient

Answer 5 of 7 · Question 9 part a.i · 0 marks

2020 JAN · Paper 02 · extract 2 of 3 · find part a.i in this mark scheme extract

Question Number	Scheme	Marks
(b)	**No value for *q***: B0B1B0B1B1 available. **Incorrect *q*:** B1B1B0B1B1 available.	
B1ft	Equations of asymptotes seen or lines parallel to axes passing through $x = 1$, $y = 3$ drawn. $y = 3$ or their q. **Must have a value for *q*.**	
B1ft	Coordinates of crossing points seen explicitly or marked on the sketch. Must have $y = 2$; may have $x = 2 / q$ (value for q not needed)	
B1	Two branches in the correct "quadrants" Must have $q = 3$ for this mark. Asymptotes drawn.	
B1ft	There must be at least one branch of the curve drawn and 2 asymptotes drawn and labelled on the diagram by showing the coords of the points where they cross the axes or with their equations. The curve must not touch (or cross) either asymptote. ft their asymptotes**, inc *y* = *q***	
B1ft	Both crossing points clearly marked on their diagram. ft their crossing points .	
(c)		
M1	Eliminate y between the line and the curve equation. May use q or their value for q	
M1	Obtain a 2 or 3 term quadratic. May use q or their value for q.	
dM1	Solve their equation to obtain 1 or 2 values of x Depends on both M marks above.	
A1cao	$x = 2$ from a correct equation. If $x = 0$ is seen it must be clear that $x = 2$ is the only answer If x is eliminated: M1 elimination M1 obtain quadratic in y M1 solve for y A1 complete to a single value of x	

Type MS · Finding unknown constants in a curve from its gradient

Answer 5 of 7 · Question 9 part a.i · 0 marks

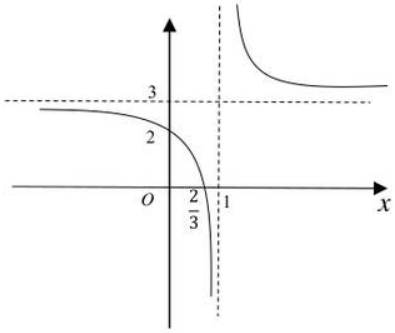
2020 JAN · Paper 02 · extract 3 of 3 · find part a.i in this mark scheme extract

Question Number	Scheme	Marks
	$\frac{dV}{}$	

Type MS · Finding unknown constants in a curve from its gradient

Answer 6 of 7 · Question 9 part a.ii · 7 marks

2020 JAN · Paper 02 · extract 1 of 3 · find part a.ii in this mark scheme extract

9(a)(i)	$\frac{2}{p} = 2$ so $p = 1$ *	M1A1cso
(ii)	$\frac{dy}{dx} = \frac{q(x-1)-(qx-2)}{(x-1)^2}$	M1A1
	When $x = 0$ $\frac{dy}{dx} = \frac{-q+2}{1} = -1, \Rightarrow q = 3$	M1A1,A1 (7)
(b)		B1ft B1ft B1 B1ft B1ft (5)
(c)	$x + 2 = \frac{3x-2}{x-1}$	M1
	$x^2 - 2x = 0$	M1
	$x(x-2) = 0$ $x = 2$	dM1A1cao(4) **[16]**

(a)(i)M1	Set $x = 0$ in the curve equation and equate result to 2. Obtain a value for p.
A1cso	Correct value of p obtained from a correct equation.
(ii)M1	Attempt the quotient rule. (formula is given on formula page). Denominator must be $(x-1)^2$. Numerator to be $q(x-1)-(qx-2)$ or $(qx-2)-q(x-1)$ **Must use $p = 1$ now or later.**
A1	Fully correct derivative
ALT	Use product rule: $\frac{dy}{dx} = q(x-1)^{-1} - (qx-2)(x-1)^{-2}$ M1 for attempt with 2 terms similar to above, either term to be correct A1 Both terms correct
M1	Set $x = 0$ in their derivative and equate to -1
A1	Correct equation
A1	$q = 3$

Type MS · Finding unknown constants in a curve from its gradient

Answer 6 of 7 · Question 9 part a.ii · 7 marks

2020 JAN · Paper 02 · extract 2 of 3 · find part a.ii in this mark scheme extract

Question Number	Scheme	Marks
(b)	**No value for *q***: B0B1B0B1B1 available. **Incorrect *q*:** B1B1B0B1B1 available.	
B1ft	Equations of asymptotes seen or lines parallel to axes passing through $x = 1$, $y = 3$ drawn. $y = 3$ or their q. **Must have a value for *q*.**	
B1ft	Coordinates of crossing points seen explicitly or marked on the sketch. Must have $y = 2$; may have $x = 2 / q$ (value for q not needed)	
B1	Two branches in the correct "quadrants" Must have $q = 3$ for this mark. Asymptotes drawn.	
B1ft	There must be at least one branch of the curve drawn and 2 asymptotes drawn and labelled on the diagram by showing the coords of the points where they cross the axes or with their equations. The curve must not touch (or cross) either asymptote. ft their asymptotes**, inc *y* = *q***	
B1ft	Both crossing points clearly marked on their diagram. ft their crossing points .	
(c)		
M1	Eliminate y between the line and the curve equation. May use q or their value for q	
M1	Obtain a 2 or 3 term quadratic. May use q or their value for q.	
dM1	Solve their equation to obtain 1 or 2 values of x Depends on both M marks above.	
A1cao	$x = 2$ from a correct equation. If $x = 0$ is seen it must be clear that $x = 2$ is the only answer If x is eliminated: M1 elimination M1 obtain quadratic in y M1 solve for y A1 complete to a single value of x	

Type MS · Finding unknown constants in a curve from its gradient

Answer 6 of 7 · Question 9 part a.ii · 7 marks

2020 JAN · Paper 02 · extract 3 of 3 · find part a.ii in this mark scheme extract

Question Number	Scheme	Marks
	$\frac{dV}{}$	

Type MS · Finding unknown constants in a curve from its gradient

Answer 7 of 7 · Question 11 part b.i · 4 marks

2022 JUN · Paper 02 · extract 1 of 4 · find part b.i in this mark scheme extract

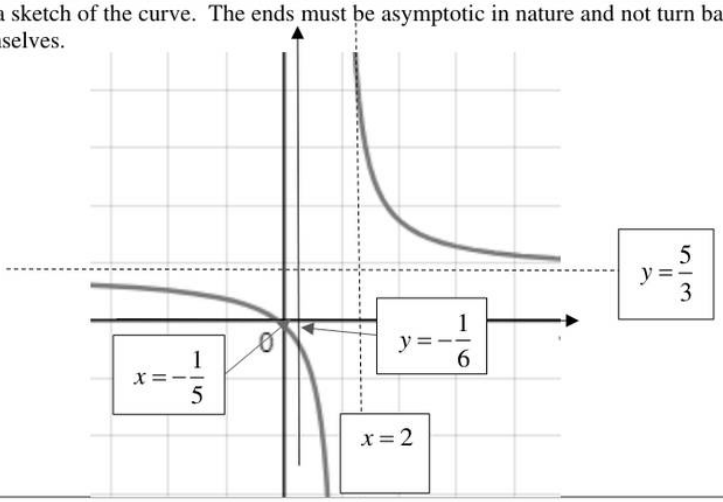
11 a	$\frac{dy}{dx} = \frac{(2a-1)(ax-6) - a((2a-1)x+1)}{(ax-6)^2}$	M1A1A1 (3)
b (i)	At $x = 0$ $\frac{dy}{dx} = -\frac{11}{12}$	B1
	So $\frac{dy}{dx} = \frac{-6(2a-1)-a}{36} = -\frac{11}{12}$	M1
	$-13a + 6 = -33 \Rightarrow a = 3$*	dM1 A1 cso (4)
b (ii)	$y = \frac{5x+1}{3x-6}$*	B1 cso (1)
c	Asymptote with equation drawn at $y = 2$	B1
	Asymptote with equation drawn at $x = \frac{5}{3}$	B1
	Correct curve drawn with two branches	B1
	$-\frac{1}{5}$ and $-\frac{1}{6}$ labelled on x and y axes	B1 B1 (5)
d	$\frac{12}{11}x - \frac{1}{6} = \frac{5x+1}{3x-6}$	M1
	$(3x-6)(72x-11) = 66(5x+1)$	M1
	$x(72x-265) = 0$ oe	M1
	$x = \frac{265}{72}$ oe	A1 (4)
		Total 17 marks

Part	Mark	Notes
(a)	**M1**	For attempting to use a correct quotient rule: The definition of an attempt is as follows: • The denominator must be correct • The terms in the numerator must be subtracted either way around • The attempt to differentiate both terms must be as follows: $(2a-1)x+1 \to \pm(2a-1)$ and $ax-6 \to \pm a$
	A1	For one term in the numerator correct
	A1	For a fully correct differentiated expression. $\frac{dy}{dx} = \frac{(2a-1)(ax-6) - a((2a-1)x+1)}{(ax-6)^2}$ No simplification is required.

Type MS · Finding unknown constants in a curve from its gradient

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	Uses Product Rule	
	M1	For attempting to use a correct product rule: The definition of an attempt is as follows: • The terms in the numerator must be added. • The attempt to differentiate both terms must be as follows: $(2a-1)x+1 \rightarrow \pm(2a-1)$ and $(ax-6)^{-1} \rightarrow \pm a(ax-6)^{-2}$
	A1	For one term correct
	A1	For a fully correct differentiation $\frac{dy}{dx} = \left[(2a-1)x+1\right](a)(ax-6)^{-2} + (ax-6)^{-1}(2a-1)$
(b)(i)	**B1**	For finding the gradient of the tangent $= -\frac{11}{12}$
	M1	For using that at the point A $x = 0$ and setting their $\frac{dy}{dx} = \left(-\frac{11}{12}\right)$ which must be the gradient of their tangent to obtain $\frac{-6(2a-1)-a}{36} = '-\frac{11}{12}' \Rightarrow \left(\frac{-13a+6}{36} = -\frac{11}{12}\right)$
	dM1	For simplifying to a linear equation and attempting to solve.
	A1	For $a = 3$ with no errors seen
(b)(ii)	**B1**	For $y = \frac{5x+1}{3x-6}$ exactly in that form. You must see evidence of substitution of $a = 3$ for this mark. There must be $y =$ …. [Condone missing $x \neq 2$]
(c)		This is a sketch of the curve. The ends must be asymptotic in nature and not turn back on themselves. 
	B1	For asymptote **with equation drawn** at $y = \frac{5}{3}$ OR $x = 2$
	B1	For both asymptotes **with equations drawn** at $y = \frac{5}{3}$ AND $x = 2$
	B1	Correct curve drawn with two branches with one branch in 1st Quadrant and another branch in Quadrants 2, 3 and 4 with the graph drawn the correct way around. The graph must be asymptotic in nature.

Type MS · Finding unknown constants in a curve from its gradient

Answer 7 of 7 · Question 11 part b.i · 4 marks

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	B1	$-\frac{1}{5}$ labelled on x-axis **or** $-\frac{1}{6}$ labelled on y- axis
	B1	$-\frac{1}{5}$ labelled on x axis **and** $-\frac{1}{6}$ labelled on y- axis
(d)	**M1**	For setting $\frac{12}{11}x-\frac{1}{6}=\frac{5x+1}{3x-6}$ [eliminating y from both equations]
	M1	For forming a quadratic equation, 2TQ or 3TQ in any order. $(3x-6)(72x-11)=66(5x+1) \Rightarrow 72x^2-265x=0$
	M1	For solving their QE provided it is either a 2TQ or a 3TQ
	A1	For $x=\frac{265}{72}$ oe

Type MS · Finding unknown constants in a curve from its gradient

Answer 7 of 7 · Question 11 part b.i · 4 marks

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